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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025 (Old Course)

Time: 03 Hours

Code: 100203 ***Subject: Chemistry*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) According to Planck's quantum theory, energy is emitted or absorbed
- (i) Continuous waves
- (ii) Discontinuously in small packets called quanta
- (iii) Randomly
- (iv) Magnetic waves

(b) In photoelectric effect, the number of photoelectrons emitted is proportional to
- (i) Intensity of incident beam
- (ii) Frequency of incident beam
- (iii) Wavelength of incident beam
- (iv) All of these

(c) The mode of excitation of molecule in IR spectroscopy is
- (i) Electronic
- (ii) Magnetic
- (iii) Vibrational
- (iv) Optical

(d) Beer-Lambert's law relates absorbance (A) with
- (i) Concentration and path length
- (ii) Pressure and temperature
- (iii) Energy and wavelength
- (iv) Volume and mass

(e) For an ideal gas, the compressibility factor (Z) is
- (i) 0
- (ii) 1
- (iii) > 1
- (iv) < 1

(f) As one moves along a given row in the periodic table ionization energy
- (i) Remains same
- (ii) Decreases from left to right
- (iii) First increases then decreases
- (iv) Increases from left to right

(g) Waterline corrosion in steel tank is an example of
- (i) Stress corrosion
- (ii) Differential aeration corrosion
- (iii) Pitting corrosion
- (iv) Differential metal corrosion

(h) Which of the following dissolved salts in water will cause the most hardness in water sample?
- (i) 10 ppm $CaSO_4$
- (ii) 10 ppm $CaCO_3$
- (iii) 10 ppm $MgCl_2$
- (iv) 10 ppm $Mg(OH)_2$

(i) Optically active stereoisomers that are not the mirror images are called
- (i) Enantiomers
- (ii) Mesomers
- (iii) Tautomers
- (iv) Diastereomers

(j) In a Diels-Alder reaction, a conjugated diene reacts with
- (i) Another diene
- (ii) A dienophile
- (iii) An alkane
- (iv) A halide ion

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Q.2 Solve both questions : **[14]**

(a) (i) State Heisenberg's uncertainty principle. What is uncertainty in the position of a particle, if the uncertainty in momentum is 10^{-11}kg m/s?
(ii) Explain why classical Newtonian and Maxwell's wave mechanics theory failed to describe the behavior of microscopic (atomic and subatomic) particles.

(b) (i) Explain the dual nature of matter and radiation. Describe de Broglie's hypothesis and briefly discuss its experimental verification.
(ii) Explain the principles of combination of atomic orbitals to form molecular orbitals.

Q.3 Solve both questions : **[14]**

(a) Draw the molecular orbital energy level diagrams for (i) N_2 and (ii) O_2 molecules. Write their molecular orbital electronic configurations, calculate their bond orders, and explain their magnetic properties.

(b) (i) What are the salient features of crystal field theory? Also explain the crystal field splitting in octahedral complexes with energy level diagram.
(ii) Differentiate between the coordination number and oxidation number of a metal ion. Further, calculate the coordination number and oxidation number of Fe in $K_4[Fe(CN)_6]$ and Ni in $Ni(CO)_4$.

Q.4 Solve both questions : **[14]**

(a) (i) State the principle of vibrational spectroscopy and derive the selection rule for vibrational transitions in diatomic molecules.
(ii) Explain the mechanism of fluorescence and give its important applications.

(b) Write informative notes on any two of the following:
(i) UV-visible spectroscopy
(ii) NMR spectroscopy
(iii) Magnetic Resonance Imaging (MRI)

Q.5 Solve both questions : **[14]**

(a) (i) Derive the Van der Waals' equation of state for n moles of gas. Explain the significance of Van der Waals' constants.
(ii) Describe the deviations of real gases from ideal behaviour and discuss their causes.

(b) (i) Explain ionic, dipolar, and van der Waals interactions with suitable examples.
(ii) Differentiate among ionization energy, electron affinity, and electronegativity, highlighting their definitions and periodic trends.

Q.6 Solve both questions : **[14]**

(a) (i) What is electronic configuration? Also give the electronic configuration of Cu, Al^{3+} and Cl^{-}.
(ii) Explain the variation of s, p, d, and f orbital energies across the periodic table.

(b) What do you understand by hybridization? Giving suitable examples explain sp^3, sp^3d and sp^3d^2 hybridization.

Q.7 Solve both questions : **[14]**

(a) (i) Predict at what temperature would the following reaction be spontaneous, given that ΔH=9.080~*kJ*/mol and ΔS=0.0357~*kJ*/(K·mol).
$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$
(ii) What is the Ellingham diagram and discuss its importance in metallurgical processes.

(b) (i) Derive Nernst equation and write its applications.
(ii) A sample of water on analysis has been found to contain the following in ppm:
$Ca(HCO_3)_2$=4.86; $Mg(HCO_3)_2$=5.84; $CaSO_4$=6.80; $MgSO_4$=8.40. Calculate the temporary and permanent hardness of the water.

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Q.8 Solve both questions : **[14]**

(a) (i) What is principle of EDTA titration? How is the permanent hardness of water is determined using EDTA method?
(ii) What are the ion-exchange resins? How will you purify water by using the resins? What are the advantages of this method over other methods.

(b) (i) What are the stereoisomers? How are they classified.
(ii) Explain the term chirality. Discuss the optical isomerism of chiral organic compounds using the example of tartaric acid.

Q.9 Solve both questions : **[14]**

(a) (i) Draw and describe the conformations of cyclohexane. Discuss the relative stability of conformers.
(ii) What are the different types of reactive intermediates involved in organic reactions?

(b) (i) Discuss elimination and substitution reactions with examples.
(ii) Give a brief note on the synthesis of aspirin.

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